

Edoardo Pierluigi Leali

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PROFILE

Master's student in High-Performance Computing within the EU-funded **European Master for High-Performance Computing** double-degree programme at Politecnico di Milano and Università della Svizzera italiana. I combine a strong engineering background with an international academic path across Italy, Switzerland, the United States and Turkey. My main interests are high-performance and heterogeneous computing, computer architecture, AI workload efficiency and data-intensive systems.

Through semester-long team projects and international study experiences, I have developed a structured working method, analytical problem solving, adaptability, communication in multicultural environments and respect for deadlines. I am motivated to join a technologically advanced company where I can contribute to complex engineering projects while continuing to learn. Open to internships and junior engineering roles in Italy or abroad, with availability for relocation and business travel.

EDUCATION

Politecnico di Milano & Università della Svizzera italiana (USI) Milan, Italy & Lugano, Switzerland
MSc in High-Performance Computing - EUMaster4HPC Double Degree 2024 – 2026 expected

- European double-degree programme focused on high-performance computing, parallel programming, heterogeneous systems and data-intensive applications.
- Relevant coursework: Parallel Computing, Numerical Analysis for Partial Differential Equations, Photonic Computing, High-Performance Data Analytics.

EUMaster4HPC Summer School Ankara, Turkey
International summer school within the European HPC programme 2025

- Attended lectures and practical activities in an international cohort, strengthening technical knowledge and collaboration in a multicultural academic environment.

Politecnico di Milano Milan, Italy
BSc in Computer Science Engineering 2019 – 2024

- Built foundations in programming, algorithms, software engineering, computer systems, databases, mathematics and engineering problem solving.

Secondary Education - Italy & United States Greater Boston Area, USA
Italian high-school diploma and U.S. High School Diploma at Arlington High School 2017 – 2019

- Completed a fourth-year exchange programme in the United States, obtaining a U.S. High School Diploma and developing strong English communication skills and adaptability.

SELECTED ACHIEVEMENTS

- **Selected for EUMaster4HPC:** admitted to a competitive European, EU-funded double-degree programme in high-performance computing, with tuition support and scholarship; part of an international cohort of around 50 students.
- **International academic profile:** studied in Italy, Switzerland, the United States and Turkey, with experience adapting to different educational systems and multicultural teams.
- **EUMaster4HPC workshops:** participated in advanced workshops on CUDA multi-node programming, FPGA programming with Intel oneAPI and MATLAB integration for high-performance computing.

SELECTED PROJECTS

AI Factory Benchmarking on MeluXina 2025
EUMaster4HPC Team Challenge - HPC Systems Luxembourg / Remote

- Designed and benchmarked AI service deployments on MeluXina, a European Tier-1 supercomputer, using SLURM-based orchestration.
- Worked on performance evaluation, workload automation, monitoring with Prometheus, metrics collection and scalability analysis.
- Translated technical measurements into practical observations on system behaviour, resource usage and deployment efficiency.

Spatiotemporal Graph Neural Networks for Wind Forecasting

2025

Course Project - Graph Deep Learning

Politecnico di Milano

- Developed data pipelines and trained graph neural-network models on real meteorological datasets for wind forecasting.
- Evaluated multiple experiments and model configurations, focusing on clear comparison of results and reproducibility.

ONN-Based Chromatic Dispersion Pre-Compensation

2024

Course Project - Photonic Computing

Politecnico di Milano

- Developed end-to-end MATLAB simulations of optical neural-network approaches for chromatic-dispersion compensation in fiber-optic communication links.
- Explored hardware-aware computing paradigms and trade-offs between signal-processing accuracy, implementation constraints and system efficiency.

Java Game Development

2023

Semester-long Team Project - Software Engineering

Politecnico di Milano

- Collaborated in a team of four to design, implement and deliver a Java game over one semester.
- Practiced task planning, version control, coordination, testing and delivery under academic deadlines.

TECHNICAL SKILLS

Programming languages: C, C++, Python, Bash, MATLAB, Java.

Parallel and high-performance computing: MPI, OpenMP, CUDA, GPU programming fundamentals, SLURM job scheduling, workload automation and scalability evaluation.

AI and data analytics: PyTorch, PyTorch Geometric, graph neural networks, data preprocessing, experimental evaluation and performance comparison.

Systems and tools: Linux/Unix environments, Git, Prometheus monitoring, LaTeX, basic scripting for automation and reproducible experiments.

Technical areas of interest: Computer architecture, memory systems, heterogeneous computing, efficient AI inference, data movement reduction and performance-oriented system design.

PROFESSIONAL STRENGTHS

- **Analytical problem solving:** used to decomposing complex technical problems and evaluating solutions through experiments and measurable results.
- **Teamwork and communication:** experienced in academic team projects, technical discussions and coordination of tasks across different backgrounds.
- **Planning and reliability:** comfortable working with deadlines, organizing project activities and documenting results clearly.
- **Adaptability:** international study experience in several countries, with exposure to different academic environments and working styles.

LANGUAGES

Italian Native

English C1 - academic and international experience, including U.S. High School Diploma

German A2

HOBBIES AND INTERESTS

- **Chess:** long-term interest that reflects strategic thinking, patience and analytical decision-making.
- **Scuba diving:** certified diver with first-aid / first-responder certification; strong focus on safety, preparation and responsibility.
- **Technology and industrial innovation:** interested in advanced computing systems and their application in industrial, automation, aerospace and semiconductor contexts.

AVAILABILITY AND MOBILITY

Available for internships, thesis collaborations and entry-level engineering opportunities. Open to roles in Milan and Italy, as well as opportunities abroad, especially in European and international environments. Flexible regarding relocation and business travel depending on the role and project.